

SANSKRITI CHOKHANI

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EDUCATION

Columbia University in the City of New York

New York, NY · Aug 2025 – Expected Dec 2026

M.S. Business Analytics

Coursework: Agentic AI, Generative AI & LLMs, Machine Learning, Applied ML, Optimization, Probability & Statistics

University of Delhi

Delhi, India · Aug 2019 – May 2022

B.S. (Honors) Statistics · GPA: 3.97 / 4.0

EXPERIENCE

OpenSesame | Data Science Apprentice, Growth

New York, NY · Jan 2026 – May 2026

- Achieved **2x** higher conversion among flagged leads by building and benchmarking **XGBoost** against L1-regularized logistic regression, using **SHAP** to surface key conversion drivers.
- Reframed channel routing and marketing budget allocation, measured by win rate and deal value, by identifying high-value account segments through segmentation analysis.

Spinny | Business Analyst, Auctions

Gurugram, India · Dec 2024 – Jun 2025

- Auction bidding model** built on price elasticity, bid distributions, and demand-supply gaps; threshold optimization experiments delivered a **45% margin increase** and **17-hour inventory turnaround reduction**.
- Fraud detection model** trained on device and geo-location signals across 11K+ records; caught high-risk dealer patterns earlier in the auction pipeline, cutting margin leakage **from 8% to 5%**.
- Cohort-based re-engagement campaign** designed and rolled out across dealer segments; reactivated **1,000+ dormant dealers** and drove **20% MoM revenue growth**.

DecisionTree Analytics | Data Analyst

Gurugram, India · Sep 2022 – May 2024

- Customer lifecycle segmentation** model built (activation, retention, churn-risk, cross-sell) across a large-scale FMCG dataset, surfacing priority segments for commercial strategy.
- SQL pipelines optimized** through indexing and partitioning, cutting query runtime **from 45 to 20 min**; automated reporting saved 5–7 hrs/week across cross-functional teams.

PROJECTS

Optimization Under Constraints: NYC Childcare Desert Elimination

Spring 2026

- Estimated the cost to eliminate NYC childcare deserts across two **Gurobi**-based integer programming models, measured by required investment under idealized and realistic scenarios.
- Quantified gap between idealized and realistic models, by testing sensitivity across imputation methods and cost assumptions.

Goodreads: Book Club Matcher Feature

Spring 2026

- Identified **unmet need** for post-reading discussion through **persona-driven user research** & app/community analysis. Prioritized 5 competing concepts via **RICE scoring** into an **MVP**-scoped agile development plan and PRD.
- Stress-tested** the plan against unvalidated behavioral assumptions, resequenced the backlog to prioritize trust-building, & drove prototype development against a defined **launch success metric** ($\geq 30\%$ opt-in rate).

Impact of Economic Variables on Fertility Rate (Published, ISBN: 978-93-91248-57-4)

May 2022

- Co-authored a regression study across 181 countries measuring causal drivers of fertility (life expectancy, education, health expenditure); validated heteroscedasticity, multicollinearity, and autocorrelation assumptions.

SKILLS

ML and Modeling: Regression, Clustering (KMeans), XGBoost, Random Forest, Lasso & Regularization, k-Fold Cross-Validation, A/B Testing, Experiment Design, Causal Inference, Feature Engineering, Time Series

Optimization: Linear, Integer and Mixed Integer Programming (Gurobi), Scenario Modeling, Sensitivity Analysis

Languages and Tools: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras), Excel, SQL, Tableau, Git/GitHub, RAG & Agentic AI (Tool Use, MCP), Web Scraping, API based Data Collection